



THE UNIVERSITY
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Fear of temperature: policy, news media and public emotions in a changing climate

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Use of AI Statement

Discussion draft updated 15 September 2026. Introduction, research questions and initial methods are drafted. The literature review, detailed evaluation, timetable and ethics sections remain in development. The submission date is provisional; some AI-use details remain to be confirmed.

Have you used AI to explore your topic? Yes. ChatGPT and Codex assisted discussions of Fear of Temperature, including its focus on social emotions and the use of computational methods to investigate them. I corrected AI interpretations that shifted the project towards a retrieval-system study or treated cultural heritage as an established premise.

Have you used AI to define your research question? Yes. AI assisted the formulation and critical review of questions about warming-related emotions, event narratives, changing topics and temporal relationships between policy and public discourse. The questions below are provisional and will be refined through source assessment and supervisory discussion.

Have you used AI to assist with your literature review? Yes. Web-assisted AI research helped locate and compare methodological and substantive literature. The accompanying research notes distinguish the sources and sections inspected from material still requiring further reading. I remain responsible for the accuracy of citations and interpretations in the submitted proposal.

Have you used AI to assist with formatting the answers presented here? Yes. AI assisted English drafting, organisation, translation and language editing. Final formatting will follow the supplied template.

Have you used AI for other aspects of your work? Yes. AI assisted technical feasibility review, research decision recording and the comparison of proposed analysis methods. This proposal revision did not collect a new corpus or run the proposed NLP and time-series experiments. Earlier project work will be disclosed from its actual records.

I acknowledge the use of generative AI tools in completing this assignment. Details of which tools were used and how they were used are provided in the table below, along with appropriate in-text and full references. I take responsibility for critically evaluating and integrating the AI-generated content, and ensuring it adheres to academic integrity standards.

AI tool and recorded use	Activities relevant to this proposal
ChatGPT; project discussions recorded on 11 and 15 September 2026; exact model versions to be confirmed from records	Topic exploration, research questions, literature assistance, methodological critique, writing and language support
Gemini; date and model version to be confirmed	Suggested analysis modules and an agenda-setting and lag-analysis direction, subsequently selected and revised by the author
Codex; 15 September 2026; exact model identification to be completed from the usage record	Material review, source checks, decision recording, feasibility review and incremental Markdown drafting

Editorial task: transfer this factual record into the template's AI-use table and add verified tool references before submission. Do not inherit example dates or model names from the template.

Contents

Use of AI Statement	2
1. Introduction	4
1.1 Motivation and significance	4
1.2 Scope and temporal framing	4
1.3 Broad aim and intended contributions	5
2. Background and Literature Review	6
3. Project Plan	7
3.1 Research objectives and questions	7
3.2 Initial methodological design	7
3.3 Further planning to be developed	10
References	11

1. Introduction

1.1 Motivation and significance

Fear of Temperature investigates social emotions associated with rising temperature and climate change. It asks how fear and related responses are expressed, what consequences and events they are connected to, and how their prominence changes across policy and public discourse. The substantive interest is in the social experience and communication of warming. Natural language processing (NLP) and artificial intelligence provide the principal means of collecting and analysing the textual evidence needed to investigate that interest at scale.

A central empirical problem is who changes first: government and policy discourse, news media reporting, or public expressions of warming-related fear and anxiety. Climate-related concern has previously been studied in relation to extreme weather, scientific information, media coverage and political cues (Brulle et al., 2012). Research on affective agenda dynamics has also examined temporal relationships between government, media and public expressions using social-media time series (Zhou et al., 2023). The project will investigate synchronisation, leading and lagging relationships, and feedback between these sources. Close alignment between government and news discourse is a working expectation to examine, rather than a uniform relationship assumed across countries or periods.

The earlier phase of Fear of Temperature used lexical inventories, dictionary-based search and n-gram analysis to explore temperature-related discourse. The proposed phase will extend this work through a newly collected natural-language corpus, structured database records, Transformer-based text representations and semantic retrieval. Three analytical components will then examine events and relations, aspect-specific sentiment and emotions, and topics and semantic patterns over time. Their outputs will support a common investigation of social emotions, with temporal analysis connecting textual patterns to independently recorded events and climate observations.

This methodological transition is intended to reduce dependence on the researcher's manual selection of expressions and individual passages. Consistent computational procedures will enable a broader body of material to be searched and analysed under documented rules. Limited, structured validation will assess what those procedures actually measure, including whether an emotional expression concerns warming, its anticipated consequences or a policy response. This distinction matters: criticism of an expensive climate policy and fear of future heat damage can both contain negative language while expressing different attitudes towards different objects.

1.2 Scope and temporal framing

The collection will prioritise English-language material associated with the United States, Europe, Australia and New Zealand. Broad geographical coverage is a research objective. Government and policy texts, news media reports, and public social-media and forum contributions form three principal source-role strata. Candidate social sources include Facebook, Reddit, YouTube comments, Bluesky, Mastodon, and selected public discussion and question-and-answer forums, including relevant Stack Exchange communities. Geography and language will primarily inform sampling, harmonisation and statistical adjustment, while the main comparison concerns source roles and groups of speakers. Such adjustment will not be assumed to remove all contextual differences. English text alone will not establish a writer's location or cultural identity.

The substantive emphasis is on anticipated warming and its future consequences. Direct high-temperature threats and longer-term expectations will be collected within the same corpus and labelled during analysis, allowing both to occur in one passage. Initial distinctions concern personal survival, family and future generations, and collective or societal anxiety. Temporal horizons will be recorded separately as immediate or near-term, longer-term anticipated, and distant or existentially imagined futures, with unspecified and overlapping cases retained. A concern for children does not automatically specify a distant horizon, and collective anxiety can concern a present emergency.

Around 1988 is the preferred starting point for the main collection. The establishment of the IPCC in that year provides an institutional anchor, while the reporting of James Hansen's congressional testimony provides a prominent communication anchor (IPCC, n.d.; Conway, 2008). These events motivate the proposed sampling boundary without establishing that a uniform public emotional transition occurred at that moment. Recoverable material from 1938 onwards may provide an earlier contextual layer, anchored by Callendar's contribution to the relationship between artificial carbon-dioxide production and temperature (Callendar, 1938).

The collection's historical span and the observation window for each analysis will be distinguished. Cross-source time-series comparisons will use periods with sufficient overlapping coverage, rather than assume that present-day social platforms provide continuous records from 1988. The analysis will focus on emotions expressed in the collected material. Claims about wider populations will depend on the sampling evidence and any available external validation. Cultural heritage may inform later contextual discussion, but it is not an assumed status of the material or a premise required by the research design.

1.3 Broad aim and intended contributions

The broad aim is to investigate the leading, lagging and reciprocal relationships between government and policy discourse, news media reporting, and public emotions concerning rising temperature, with particular attention to anticipated futures, policy events and the attribution of causes and responsibilities.

The first intended resource contribution is an openly accessible corpus and database release, developed through documented online collection and source verification. The release will include the text, metadata, identifiers and derived material that can be shared under the relevant source conditions. A modular research architecture will connect collection, representation, extraction, emotion analysis, topic analysis and temporal modelling. Algorithmic or methodological improvements may emerge where the empirical work demonstrates a need and provides evidence of their value. The principal scholarly outcome will be a thesis grounded in substantive findings, with the ambition of developing results suitable for publication.

2. Background and Literature Review

Next writing stage. Planned coverage: social emotions and climate communication; agenda-setting and temporal relationships; event and relation extraction; aspect-specific sentiment and emotion recognition; dynamic topics and semantic change; and the specific contribution in relation to the closest studies. The previous retrieval-centred literature review will be selectively reused, not transferred as the organising argument.

3. Project Plan

3.1 Research objectives and questions

The first empirical objective is to establish which source roles change first in their warming-related attention, emotional expression and interpretive frames, and whether major policy events coincide with changes in their levels, trends or relationships. Event extraction, emotion analysis and temporal topic analysis will provide complementary evidence.

RQ1 - Who changes first? What leading, lagging, synchronous or reciprocal relationships are observable between warming-related government and policy discourse, news media reporting, and public fear and anxiety over periods of comparable coverage?

RQ2 - Changes around policy events. Around independently dated policy and institutional events, do public fear, anxiety and semantic framing exhibit immediate changes in level or subsequent changes in trend, and how do these changes relate to government and media discourse before and after the event?

RQ3 - Attribution and anticipated futures. How do government, media and public texts connect warming-related fears to events, time horizons, causal mechanisms and responsible actors, and how do these attributions change through the adoption, contestation or reinterpretation of policy frames?

The supporting objectives are to:

1. Collect and document an English-language corpus with broad geographical coverage, construct a database linking text to its source and temporal context, and prepare an open release within the sharing conditions of the contributing sources.
2. Implement and evaluate all three required analytical components: event and relation extraction, aspect-specific sentiment and emotion analysis, and temporal topic and semantic analysis.
3. Construct role-specific measures of attention, emotional expression and attribution, and evaluate their lead-lag relationships and event-associated level and slope changes using independent climate and event records.
4. Produce substantive findings for the thesis and a reproducible research architecture, assessing any proposed algorithmic improvement against an appropriate baseline before claiming a separate methodological contribution.

3.2 Initial methodological design

Corpus construction and representation

Online collection will retain natural-language text together with provenance, source role, available geographical evidence and relevant dates. The database will distinguish publication dates, events mentioned in a text and collection dates. Transformer-based encoders will generate vector representations for semantic retrieval and downstream analysis; a vector index will remain linked to the original text records. Retrieval will identify candidate passages for analysis, with surrounding context retained to resolve emotional targets, attributed statements and event references.

Source role, publication channel, quoted speaker and people described in a passage will be recorded separately. A government account on a social platform remains a government source; a news report

quoting a citizen retains both the media publication role and the attributed speaker. Further public-group categories will require observable, documented evidence rather than inferred private identities. Texts will be encoded within a common, versioned representation space and then grouped by role, time and supported contextual attributes. Separate unaligned vector spaces will not be trained for each group or event window. Reproduced releases, syndicated news and reposts will be identified so that shared wording does not automatically count as independent emotional convergence.

The source assessment will record coverage across the United States, European settings, Australia and New Zealand. It will consider Facebook and Reddit discussions, comments responding to relevant YouTube videos, public Bluesky and Mastodon posts, and topic-based or regional forums. Sustainable Living and Earth Science Stack Exchange are candidates for a supplementary question-and-answer stratum. These candidates serve different potential roles: event-linked comments, conversational narratives, short public reactions and questions about climate consequences or responses. Their suitability and relevant emotional content will be assessed through pilot sampling.

Specific communities, policy repositories, access routes and time windows will be selected after examining availability and comparability. Public visibility, programmatic access and permission to redistribute text will be assessed separately. Sources requiring approved research access will remain conditional until that access is established. The open corpus release will contain material whose redistribution conditions permit inclusion; access-restricted analysis data will be handled under their respective conditions. A broad collection will not by itself establish representative national samples. Cross-region results will therefore identify the sources and populations they can support.

Three required analytical components

Event and relation extraction will identify participants, actions or relations, objects, time and place, and connect these elements to warming-related consequences and responses. Named-entity recognition and dependency parsing may supply components of this process, alongside relation extraction. Extracted links will represent claims or associations made in the text; a narrative that links warming to migration will not by itself establish that causal relationship in the world.

Semantic role labelling will identify predicate-argument structures (Palmer et al., 2005), supplemented by emotion-cause pairing and contextual relation extraction (Xia & Ding, 2019). The attribution design has two axes: temporal and event attribution, and causal and responsibility attribution. The first links expressions to mentioned events and temporal references. The second distinguishes explanatory mechanisms, actors assigned blame, and actors assigned duties to respond. Candidate frames include natural processes or uncontrollability, industrialisation, institutional failure, individual consumption and collective human responsibility. These categories may coexist and will not be assigned predetermined historical periods. Explicit, implicit, negated, uncertain and unexpressed attribution will be distinguished; an explicit causal phrase is not required for inclusion. Semantic roles alone do not establish culpability, and the procedure will retain evidence spans for each extracted link.

Aspect-specific sentiment and emotion analysis will distinguish the object of an attitude from its polarity and emotional category. Aspect-based sentiment analysis provides a task formulation for identifying what an evaluation concerns (Pontiki et al., 2014). Fine-grained emotion resources such as GoEmotions provide candidate categories and models (Demszky et al., 2020), but require assessment on the project's sources. Fear, anxiety, grief and opposition to a policy will not be combined into a single score by assumption. The analysis will check whether the emotion concerns

warming itself, an expected consequence or a proposed response, and whose emotion is being expressed or reported.

Threatened people or social units, temporal horizons, and attribution categories will be stored as distinct, potentially overlapping labels. Government and media texts may describe risk or report others' emotions without expressing fear themselves. Risk framing and expressed emotion will therefore produce separate measures. Policy agreement or scepticism will also be distinguished from emotion: a person may endorse a policy while fearing its insufficiency, or reject its framing while expressing substantial anxiety.

Temporal topic and semantic analysis will examine changing topics and expressions using Transformer representations and a topic-modelling approach such as BERTopic (Grootendorst, 2022). Topic prevalence and representative passages will be compared across time and source groups. Claims about changing word or concept meanings will require contextual evidence and controls for source composition; changes in topic frequency alone will not be treated as proof of semantic change. The precise semantic-change procedure remains to be specified in the next methods increment.

These components will share document identifiers and contextual records so that a temporal change in emotional expression can be examined alongside its associated events, topics and passages. A stratified validation sample will assess extraction quality, emotional targets and classification errors across sources and periods. The main corpus analysis will use documented computational rules to reduce discretionary selection of evidence.

First empirical analysis: attention and temporal relationships

The initial measures will separate the proportion of comparable source material discussing warming from the proportion of warming-related material expressing a specified emotion. If collection provides only climate-related texts, the analysis will report emotional composition within that collection rather than infer warming's share of all platform discussion. Embedding similarity will measure semantic relatedness; it will not automatically serve as an anxiety intensity score.

Textual measures will be aggregated by source role and time and aligned with independently compiled scientific and policy event dates and suitable climate observations. Global temperature anomalies and local heat exposure address different questions and will be distinguished. Public search-interest data may supplement the comparison where coverage permits. The aggregation interval and common analysis period will follow the temporal resolution and completeness of the source records. Geography, platform and other documented compositional differences will be addressed through stratification, weighting or model terms as appropriate; original scales and subgroup coverage will remain available for interpretation.

Cross-correlation will explore leading and lagging associations after addressing trends, seasonality and serial dependence. A parsimonious vector autoregression and Granger tests are planned where observation counts, stationarity and model diagnostics support their use. The tests will assess additional predictive information conditional on the included series, with their causal limits made explicit (Shojaie & Fox, 2021). Event-based dynamic regression is a candidate for sparse scientific or policy interventions. Temporal associations will be interpreted in light of shared drivers, feedback, measurement error and changes in source coverage. Any analysis of public emotion preceding policy action will require a separately defined policy-response measure.

Policy events, changes in trend and responsibility frames

Policies will serve as both dated events and texts. Initial candidate anchors include the establishment of the IPCC in 1988, adoption of the Kyoto Protocol on 11 December 1997, and adoption of the Paris Agreement on 12 December 2015 (IPCC, n.d.; UNFCCC, n.d.-a; UNFCCC, n.d.-b). Institutional establishment, adoption, signature, entry into force and media reporting will be recorded as different event types or dates. The precise operational date for the 1988 institutional event remains to be established. Historical candidates will be included in an analysis only where appropriate before-and-after material exists; early observations will not be represented as social-platform data.

An initial window of six months before and six months after each eligible event will support examination of emotional and semantic trajectories. Repeated observations within the window will be used to distinguish an immediate level change from a change in the subsequent growth or decline rate. Segmented interrupted time-series models provide a starting point for this distinction (Lopez Bernal et al., 2017). Twelve monthly observations would provide limited information for a complex model: finer aggregation requires adequate source coverage, and a longer baseline may be needed for seasonality and trend estimation. Window sensitivity, anticipatory discussion, overlapping events, serial dependence and source changes will be examined. Effect sizes and uncertainty will be reported whether or not estimates meet a significance threshold.

A dated event does not by itself provide the assumptions needed for causal regression discontinuity in time (Hausman & Rapson, 2018). Negotiations and media coverage may precede an adoption date, and contemporaneous events may affect public expression. Causal interpretation will require a defensible counterfactual, supported where feasible by appropriate comparison series, pre-event trajectories and placebo dates. Otherwise, the result will be described as an event-associated change rather than proof that a policy altered public fear.

Before-and-after vector distributions will be compared within the same representation space. Policy passages will supply reference material for examining the presence and transformation of responsibility frames, alongside independently defined warming and fear concepts. Proposed measures of semantic concentration or frame proximity will be specified and validated before substantive testing. Similarity to policy language will be interpreted together with stance, attribution and quotation evidence, since repeating a frame can express agreement, criticism or reporting. Where many passages come from the same publisher, author or time interval, uncertainty estimation will account for that dependence rather than treat every passage as an independent event observation.

This design connects the two uses of policy: event timing helps locate changes, while textual analysis examines how causes, blame and duties are allocated and subsequently reproduced or challenged. The extracted attribution is a property of discourse; it does not assign actual responsibility for climate change or establish who caused a change in public emotion.

3.3 Further planning to be developed

Subsequent increments will specify source selection and access, sampling and database fields, module-specific evaluation, the temporal model protocol, milestones, resource requirements, risks and ethics. These require source assessment and the actual course and thesis timetable. The current objectives and design are proposals, not completed experiments or final parameter choices.

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